--- Part A: Theory ---

1. What is Statistical Distribution?

A **Statistical Distribution** describes how the values of a variable are spread or distributed.

It shows:

- Possible values of the variable
- Probability of each value
- Shape of the data (normal, skewed, uniform, etc.)

🔹 **Why Important?**

- Helps understand data behavior
- Used for prediction and inference
- Forms the base of statistical analysis

Example:

- Heights of students → Usually follow **Normal Distribution**
- Number of customers arriving per hour → May follow **Poisson Distribution**

---

2. What is a Q-Q Plot and Why is it Used?

A **Q-Q (Quantile-Quantile) Plot** compares:

- Quantiles of your dataset
- With quantiles of a theoretical distribution (like Normal)

🔹 **Purpose:**

- To check if data follows a specific distribution (usually Normal)
- To detect skewness or outliers

🔹 **Interpretation:**

- Points on straight line → Data follows distribution
- Curved pattern → Skewed data
- Extreme deviation → Outliers present

3. Difference Between Discrete and Continuous Distributions

Feature	Discrete Distribution	Continuous Distribution
Data Type	Countable values	Infinite values in range
Example	Number of students	Height, weight
Probability	P(X = x) possible	P(X = exact value) = 0
Function	PMF	PDF

Example:

- Discrete → Dice roll
- Continuous → Temperature

---

4. What is Bernoulli Distribution?

A **Bernoulli Distribution** is a discrete distribution with only **two outcomes**:

- Success (1)
- Failure (0)

Formula:

$P(X=1) = p$
$P(X=0) = 1 - p$

Example:

- Tossing a coin (Head = 1, Tail = 0)
- Pass/Fail exam

It is the simplest probability distribution.

---

5. What is Binomial Distribution?

A **Binomial Distribution** represents:

- Number of successes in n independent Bernoulli trials.

Conditions:

- Fixed number of trials (n)
- Two outcomes (success/failure)
- Constant probability (p)
- Independent trials

Formula:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Example:

- 10 coin tosses → probability of getting 6 heads

---

6. Log-Normal Distribution

A variable follows **Log-Normal Distribution** if:

$$\ln(X) \sim Normal$$

This means:

- The logarithm of X is normally distributed.
- Data is positively skewed.

Characteristics:

- Right-skewed
- Cannot be negative
- Heavy upper tail

Example:

- Income distribution
- Seller revenue in e-commerce

---

7. Power Law Distribution

A **Power Law Distribution** follows:

$$P(X) \propto X^{-\alpha}$$

Characteristics:

- Extremely right-skewed
- Very heavy tail
- Few very large values

Example:

- Social media followers
- Wealth distribution
- Website traffic

Difference from Log-Normal:

- Power-law tail is heavier
- Extreme values more frequent

---

8. What is Box-Cox Transform?

The **Box-Cox Transform** is used to:

- Make skewed data more normal
- Stabilize variance

Formula:

$$Y(\lambda) = \begin{cases} \dfrac{X^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln(X), & \lambda = 0 \end{cases}$$

Why Use It?

- Improve regression performance
- Reduce skewness
- Make Q-Q plot more linear

---

9. Poisson Distribution (With Example)

The **Poisson Distribution** models:

- Number of events in fixed interval of time/space
- Events occur independently

Formula:

$$P(X = k) = \frac{e^{-\lambda}\lambda^{k}}{k!}$$

Where:

- λ = average number of events

Example:

- Customers arriving in 1 hour
- Number of calls in a call center

If average calls = 5 per hour
What is probability of exactly 3 calls? → Use Poisson formula.

---

10. What is Z-Score Probability?

A **Z-score** measures how many standard deviations a value is from the mean.

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = value
- μ = mean
- σ = standard deviation

Purpose:

- Convert normal distribution into standard normal
- Find probability using Z-table

Example:

If Z = 2 → Value is 2 standard deviations above mean.

11. Difference Between PDF and CDF

Feature	PDF (Probability Density Function)	CDF (Cumulative Distribution Function)
Meaning	Density at a point	Probability up to a value
Used For	Continuous variables	Both discrete & continuous
Value Type	Can be >1	Always between 0 and 1
Relationship	Integral of PDF = CDF	Derivative of CDF = PDF

Concept:

- PDF → "Height of curve"
- CDF → "Area under curve"